Witnesses, Not Amplifiers:
What Specialist Agents Contribute to Epistemic Quality

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Abstract

Multi-agent pipelines are often built on the premise that specialist models bring better judgement to their domains, and that an aggregator's job is to preserve it. We tested that premise directly and found both halves wrong. A domain-specialised medical model, asked a question under a neutral prompt, qualifies its claims no more than a generic model of similar size—0.15 against 0.14. Its apparent richness inside a pipeline comes entirely from the instruction in its system prompt. And a single well-prompted model matches an entire three-specialist council for sheer quantity of qualification (0.50 against 0.57, overlapping intervals). On this measure, specialists contribute nothing that an instruction does not.

Worse, tuning them upward is actively harmful. Prompting or fine-tuning a specialist to qualify more than doubles what it writes, and the finished answer gets *worse*—1.21 untrained against 0.92 for both methods. Extending the treatment to more specialists makes it worse monotonically ($\rho = -0.80$). The aggregator does not preserve what arrives; it decides afresh, and reads saturated hedging as noise to be edited out. What specialists actually contribute is narrower and more useful: *grounds* for a conditional instruction. The council was the only arrangement we tested that qualified heavily where warranted and not at all where it was not (0.00 $[0.00, 0.00]$ against a lone prompted model's 0.15 $[0.08, 0.22]$), because its final instruction can check what the specialists flagged. A solitary model has nothing to check. Specialists are worth their cost as witnesses, not as amplifiers—and the practical consequence is that effort spent making them more careful is worse than wasted, while effort spent on the aggregator's instruction is the only thing that pays.

1 Introduction

The case for multi-agent pipelines usually rests on division of labour: a medical model should reason better about medicine, a legal model about law, and an aggregator should combine their contributions without losing what makes each valuable. Where the valuable thing is epistemic—a flagged assumption, a noted uncertainty, a caveat about currency of information—the same case implies two things. Specialists should be better at producing it, and the aggregator's job is to preserve it.

We set out to strengthen a pipeline along exactly those lines and could not. This paper reports what we found instead, in the order that makes the argument rather than the order we found it.

Neither premise survives contact with a control. Specialisation does not produce epistemic care: a domain-tuned medical model and a generic model of similar size, given the same neutral prompt, qualify their claims at indistinguishable rates. The specialist looks careful inside a pipeline only because its system prompt tells it to be. And quantity of care does not require specialists at all—one model with a good instruction matches a full council.

The second premise fails harder, and in the opposite direction from the obvious failure mode. Making specialists more careful does not merely fail to help; it makes the finished answer worse, reliably, by two different methods and monotonically in the number of specialists treated. An aggregator does not carry qualification through. It decides how much to write, largely independently of how much it received, and past a point additional input suppresses rather than adds.

What remains is a narrower role, and we think a correct one. A specialist's contribution is not volume but *evidence*: a record of what was flagged, against which the aggregator's instruction can be made conditional. That distinction is what separates the council from a lone prompted model. The two match on how much they qualify; they differ completely on whether they qualify when nothing warrants it. One can check what its specialists said. The other is guessing.

We call this the difference between a witness and an amplifier, and the rest of the paper is the evidence for it.

2 Setup

Pipeline. Four models running locally on one 32 GB machine: a planner/aggregator (Phi-4-14B unless stated) and three domain specialists (Llama3-Med42-8B, Saul-7B-Instruct, Qwen-Open-Finance-8B), all quantised. The planner decomposes a question and sends each specialist a self-contained sub-question; specialists never see the original query. The aggregator then writes the final answer under an instruction that asks it to surface disagreements and to preserve specific things the specialists flagged. The full instruction is in Appendix A.

What we measure. We count four families of epistemic behavior per 1,000 characters: disclosing that information may post-date training, labelling a number as an assumption, distinguishing jurisdictions or regimes, and calibrated hedging. A fifth—distinguishing near-synonymous terms of art—fell below detection and is dropped. Counting is done by pattern matching, cross-checked on the load-bearing comparisons by a calibrated entailment model and by blinded pairwise LLM judges shown both texts in each order (Appendix B).

Questions. Seven cases spanning healthcare, law and finance. Six warrant qualification: they turn on guidance that may have changed, on quantities the question leaves unstated, or on conflicting jurisdictions. The seventh deliberately warrants none—every figure it needs is given and nothing in it has moved in a decade. That case does most of the work in this paper, because a system that qualifies everything is otherwise indistinguishable from one that qualifies the right things.

Scale. Roughly 1,300 runs, five seeds per condition, intervals by bootstrap. Predictions were registered before the runs that test them, and we report the ones that failed.

3 Specialisation does not produce epistemic care

The first premise is testable in isolation. Take the specialists out of the pipeline, give them a neutral instruction—answer the question directly and substantively—and see whether they qualify more than a generic model.

They do not. Med42-8B, a Llama-3 model tuned on clinical data and preference- aligned, produces 0.15 $[0.04, 0.28]$. Mistral-7B-Instruct produces 0.14 $[0.04, 0.26]$; Qwen2.5-7B-Instruct 0.14 $[0.06, 0.24]$; a 20B mixture-of- experts, measured the same way, 0.14. Four models on four different architectures, spanning clinical fine-tuning and generic instruction tuning, all sit within 0.01 of each other.

The same model, placed in the pipeline as a specialist, produces 1.31 $[1.02, 1.63]$—nearly nine times more,

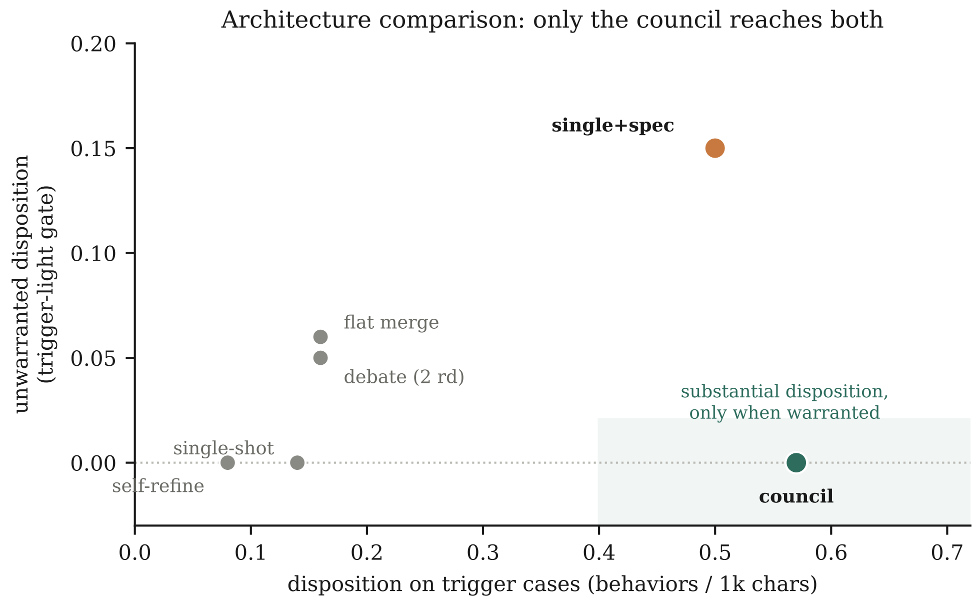

Figure 1: Six arrangements, one model in every role (210 runs). Horizontal: qualification where it is warranted. Vertical: qualification where it is *not*. Three specialists merged plainly, and two rounds of debate, barely leave the single-model corner. A lone prompted model reaches full volume but qualifies a question that warrants nothing. Only the council reaches the shaded region.

with intervals nowhere near overlapping. Nothing about the model changed. What changed is that its system prompt in the pipeline says *"Flags training-cutoff uncertainty EXPLICITLY when guidelines or evidence may have evolved,"* and that the planner appends a further reminder to do so when it judges the question time-sensitive—which it did in 30 of 34 runs.

So the specialist's epistemic richness is an instruction, not an expertise. This is worth stating plainly because it is the premise on which specialist pipelines are usually justified, and because it is cheap to check and, as far as we can tell, rarely checked.

4 Nor do specialists supply quantity

If specialists do not produce care by being specialists, perhaps they produce it by being many. We tested this by holding the model fixed—the same 20B model in every role—and varying only the shape of the arrangement across six designs, so that any difference is attributable to structure alone (Fig. 1).

Three specialists whose answers are merged with a plain instruction reach 0.16 $[0.11, 0.22]$—indistinguishable from one model answering alone at 0.14 $[0.09, 0.19]$. Two full rounds of debate, in which each specialist revises after reading the others, reach 0.16 $[0.12, 0.20]$: more deliberation, no more care. A single model asked to draft, critique itself and revise reaches 0.08, the lowest of any arrangement—self-criticism tightens prose and the qualifications are the first thing to go.

Meanwhile a single model given a good instruction and no specialists at all reaches 0.50 $[0.41, 0.60]$, against the full council's 0.57 $[0.45, 0.71]$. The intervals overlap. On quantity of qualification, three specialists and a planner buy nothing over one model and a paragraph of instruction.

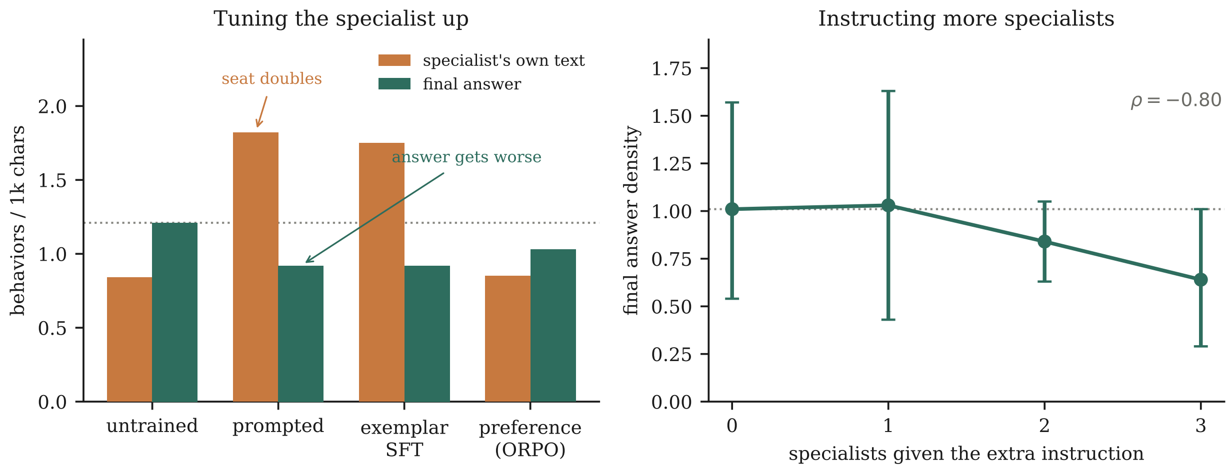

Figure 2: *Left:* treatments that raise what a specialist writes lower what the pipeline finally says; the dotted line is the untrained final answer. *Right:* extending the treatment from zero to three specialists lowers the final answer monotonically.

Table 1: The legal specialist under four treatments: what the specialist writes, and what the pipeline finally says (bootstrap 95% intervals over $n = 30$ runs each).

Treatment	Specialist's own text	Final answer
untrained	0.84 $[0.62, 1.06]$	$\mathbf{1.21}$ $[0.93, 1.49]$
prompted	$\mathbf{1.82}$ $[1.38, 2.30]$	0.92 $[0.67, 1.19]$
exemplar fine-tuning	$\mathbf{1.75}$ $[1.45, 2.08]$	0.92 $[0.72, 1.13]$
preference optimization	0.85 $[0.58, 1.15]$	1.03 $[0.81, 1.28]$

5 Tuning specialists upward makes the answer worse

The remaining hope for the standard picture is that specialists are simply under-tuned—that a specialist trained to qualify more would push more through. We tested this at the specialist in three ways, and it fails in a way that is more informative than a null (Fig. 2, left).

Prompting the specialist more than doubles what it writes, from 0.84 to 1.82. Fine-tuning it on qualification-rich examples does the same, 1.75. Both are real: the intervals exclude the untrained baseline, and the specialist's text is visibly more careful. Both make the finished answer *worse* than leaving the specialist alone—0.92 against 1.21.

Extending the treatment across specialists confirms the direction. Giving the extra instruction to zero, one, two and then all three specialists produces final answers of 1.01, 1.03, 0.84, 0.64: a monotone decline, $\rho = -0.80$. Care does not accumulate across specialists and then get diluted. It is not accumulating at all.

The texts suggest why. A specialist told to flag uncertainty everywhere flags it in every clause, and the result reads less like carefulness than like a verbal tic. The aggregator, whose instruction asks it to write a clean integrated answer, treats it accordingly and edits it out. A useful way to put this: an aggregator is not a channel that passes qualification through with some loss. It is a writer that decides how much to write, and evidence of over-qualification upstream is a reason for it to write less.

Table 2: The two effects, separated. “Volume” is qualification on questions that warrant it; “discrimination” is its absence on the question that does not.

Arrangement	Volume	Unwarranted qualification
one model, no instruction	0.14	0.00
three specialists, plain merge	0.16	0.06
one model, standing instruction	0.50	0.15
three specialists, conditional instruction	0.57	**0.00**

6 What specialists actually contribute

Return to the comparison that matters. A lone prompted model and the full council are indistinguishable on how much they qualify where qualification is warranted—0.50 against 0.57. They are not remotely alike on the question that warrants none. The lone model qualifies it at 0.15 $[0.08, 0.22]$; the council at 0.00 $[0.00, 0.00]$, on every one of five runs. The intervals are disjoint.

The reason is in the form of the instruction rather than its content. The council’s aggregator is told, in effect, *if a specialist flagged an assumption, label it as one; if a specialist flagged that information may have aged, carry that through.* Every clause is conditional on something a specialist did. When the specialists flag nothing, the conditions are not met and nothing fires. The lone model is given the same requirements as standing instructions, because there is no upstream for them to refer to, so they fire whether or not anything warrants them. It has learned the form of epistemic care with none of the judgement: asked a routine question about hybrid-work communication, it opened with a section headed “Current State Snapshot (as of my training data)” and closed with “actual results may vary.”

This is what specialists are for. Not to produce more qualification—the aggregator’s instruction sets that, and the specialists barely affect it—but to produce a record of what was actually uncertain, against which a conditional instruction can be evaluated. They are witnesses. Their value is that they can be checked, and it is destroyed if they are trained to say everything is uncertain, because a witness who always says the same thing carries no information.

Table 2 makes the decomposition explicit. Volume comes from the instruction: it is what separates rows three and four from rows one and two. Discrimination comes from having specialists *and* making the instruction depend on them: it is what separates row four from row three. Neither ingredient produces it alone.

7 Why this cannot be trained in

The obvious objection is that we simply instructed where we should have trained. We tried training, at both ends and by three methods, and it does not work at either.

At the specialist, preference optimization installs no additional qualification at all—0.85 against an untrained 0.84—and tripling the training data changes nothing, even though the larger set raised the model’s accuracy at distinguishing good examples from bad from 0.48 to 0.94. The preference was learned and not acted on. Replication on a second specialist of a different lineage produced no effect; on a third, an effect we had registered in advance failed to appear.

At the aggregator—the position this paper says decides everything—training also fails. Fine-tuning it on qualification-rich synthesis examples left its output where it started, 0.89 against 1.03 untrained. We then repeated this with the reward corrected in every way we could think of: sampling the aggregator's own outputs rather than a fixed corpus, and scoring them on the finished answer for qualifying heavily where warranted and not at all where not. It moved nothing, 0.60 against 0.56.

What did survive training is telling. After fine-tuning the aggregator on precisely the behavior its instructions elicit, the instructions still worked at full strength—the effect of adding clauses was $1.58\times$ on the trained model against $1.82\times$ untrained. If instructions and weights were competing routes to the same behavior, training on that behavior should have absorbed some of the instruction's job. It did not, which suggests they act on different things: the weights on what a model tends to do, the instruction on what it does this time.

We are careful about the strength of this claim. Our training was small— low-rank adapters over tens of preference pairs, against instruction-following that model builders installed with vastly more data. "A small amount of training did not overwrite a deeply installed capability" is weaker than "training cannot do this," and it is the one we can support. What we can say without qualification is that at every scale available to us, the instruction was free, immediate and effective, and the training was none of those things.

8 Limitations

Of five behaviors we set out to count, one fell below detection, so we report four; the effect of instructions is clear in combination but not for any single behavior alone. Pattern matching misses paraphrase and rewards brevity; two further instruments agree with it on the load-bearing comparisons, but all three are automatic and none is anchored to human judgement.

Everything ran at or below 20B parameters on one machine, with five seeds and seven questions we wrote ourselves. A frontier model as aggregator could behave differently, and the question that warrants no qualification carries a great deal of weight for something measured over five runs. The architecture comparison uses one model family in every role and one implementation of each design.

We also report defects in our own record rather than repairing them. One model we intended as a comparison produced degenerate output—26 of 30 answers were fragments ending mid-sentence—which we found only after computing a figure from them; that arm is withdrawn, and with it a comparison between alignment lineages we had intended to make. In our earliest arms, five runs per question are in fact one run from an earlier session plus four from a later one, and 133 runs record a batch label where a model identifier belongs. None of this changes a reported interval, but it weakens those arms' internal consistency.

9 Conclusion

Specialist agents in a pipeline are usually justified by what they know and defended by how carefully they say it. Neither holds up here. A domain-specialised model is no more epistemically careful than a generic one of similar size until it is told to be, and once several of them are told to be, the finished answer gets worse rather than better. A single well-prompted model matches a full council for how much it qualifies.

But the council was the only arrangement that knew when to stop, and that is not a small thing—it is the difference between a system whose caution means something and one that hedges reflexively. It came from an instruction that could check what the specialists had actually flagged, which is the one thing a solitary model cannot do at any level of prompting.

So the role specialists play is real but narrow. They are witnesses. What they are for is being checkable, and the way to ruin them is to train them to qualify everything, because a witness who always says the same thing tells you nothing. Build them to flag what is genuinely uncertain and no more. Put the effort into the aggregator's instruction, make every clause of it conditional on what the specialists said, and stop fine-tuning components for epistemic quality, which in our hands never reached the page.

Reproducibility. Code, prompts, all runs, pre-registrations, protocol amendments, refuted predictions, and an integrity audit that withdrew one arm: `https://github.com/srbobo/council-of-experts-research`.

A The aggregator's instruction

The aggregator writes in two steps—first identify tensions between the specialists, then write the integrated answer. The second step carries five requirements, of which three are the preservation clauses. Verbatim:

> **1.** Acknowledge the tensions you just identified—not abstractly, but at the points where they bite the answer.
>
> **2. PRESERVE numeric framing**—if a specialist labeled a number as a modeled assumption, your synthesis MUST also label it as an assumption (use "modeled at," "assumed"). Never adopt a specialist's modeled number as a fact.
>
> **3. PRESERVE precise vocabulary**—if a specialist used a specific legal term, jurisdiction, FDA pathway, or clinical concept, preserve it; do not smooth out precision in the name of accessibility.
>
> **4. PRESERVE caveats**—if a specialist flagged training-cutoff uncertainty or data fragility, propagate that flag into your synthesis.
>
> **5.** Use whatever structure best fits the user's question.

Clauses 2–4 are conditional on something a specialist did; clause 1 is not, and when tested alone it changes nothing. Isolating the clauses shows each lifting the behavior it names and no other, which is why we describe them as specifications rather than as general encouragement to be careful.

B Instruments

Pattern matching over four behavior families is the primary measure. Two independent instruments check the load-bearing comparisons: an entailment model calibrated against our own construction-labelled pairs, and blinded pairwise LLM judges shown both texts in each order and required to quote the evidence for their verdict. Where all three agree we report the result; where they disagree we report the disagreement. One finding did not survive—an apparent *increase* in hedging on the finance specialist appeared under pattern matching alone, and neither other instrument saw it. We retired it as an artifact of the measure.